



Class Project Phase 1

Group: 1

Team Members:

John Bostater

Tristan Andrade

Austin Mayhew

Ryan Clark

Juan Rosas Jr.

Project Overview

Arizona Medical Online Portal

Introduction

The Arizona Medical Online Portal is an innovative solution designed to simplify and enhance the interactions between healthcare providers and patients. This comprehensive digital platform allows for seamless access to medical services and information, significantly improving the efficiency of healthcare operations and enhancing patient satisfaction.

Key aspects

- **Welcome Screen and User Selection:** Upon entering the portal, users are greeted with a choice between patient and staff login, directing them to the appropriate interface based on their role. This initial segregation ensures that the user experience is tailored from the first interaction.
- **Patient Registration and Login:** Patients can create new accounts or log in using existing credentials. This screen is crucial for capturing and verifying patient information, ensuring that the data entered is secure and private.
- **Staff Selection:** Once logged in as staff, users must select whether they are doctors or nurses. This step further refines the access level, ensuring that sensitive patient information is only accessible to authorized personnel.
- **Patient Dashboard:** After login, patients access a dashboard where they can schedule appointments, view past medical records, or communicate with their healthcare providers. This centralized hub makes it easy for patients to manage all aspects of their healthcare.
- **Staff Dashboard:** Healthcare providers have access to a dashboard where they can update patient records, manage appointments, and review patient histories. This interface supports efficient management of patient care and operational tasks.
- **Security Features:** Each interaction within the portal is protected through end-to-end encryption, and data integrity checks are routinely performed to ensure that the information remains secure and unaltered.

Functional Requirements

Our team has developed a user-centric portal that includes several key functionalities aimed at improving the overall healthcare experience. Key features include a secure login system for both patients and staff, which ensures that all user interactions within the portal are safe and private. Patients can easily create and manage their accounts by providing basic personal information, such as their first name, last name, and date of birth.

Once logged in, patients have access to a range of services including the ability to schedule appointments, view past visit histories, access their medical records, and communicate directly with their healthcare providers through a built-in messaging system. This ensures that patients have all the necessary tools at their disposal to manage their health effectively.

For healthcare staff, including doctors and nurses, the portal provides differentiated access levels to ensure sensitive patient data is handled securely. Staff members can view and update patient information, manage visit records, and respond to patient messages. The design of the staff interface is intuitive, allowing for easy navigation and management of patient data, which is crucial in a fast-paced medical environment.

Non-Functional Requirements

In addition to functional capabilities, the Arizona Medical Online Portal has been designed with a strong focus on non-functional requirements to ensure a robust, reliable, and user-friendly system. The portal is designed to be highly usable, with a clear and intuitive user interface that is accessible on various devices including desktops, tablets, and smartphones. It is optimized for performance, with fast loading times and the capability to handle a high volume of concurrent users without any degradation in performance.

Reliability is a cornerstone of the portal's design, ensuring high availability and minimal downtime. Our system includes regular backups and a robust disaster recovery strategy to safeguard against data loss. Scalability is also a key consideration; the portal is built to accommodate growth in user numbers and data volume without the need for significant redesign.

Security measures are rigorously implemented, including regular updates and patches to protect against vulnerabilities and strong authentication and authorization mechanisms to prevent unauthorized access.

Conclusion

The Arizona Medical Online Portal represents a significant advancement in the digital transformation of healthcare services in Arizona. By providing a secure, efficient, and user-friendly platform, we empower patients to take control of their health and enable healthcare professionals to deliver higher quality care. This portal not only streamlines healthcare management but also ensures that both patients and providers have access to the necessary tools to ensure effective communication and management of health services. As healthcare continues to evolve, the Arizona Medical Online Portal will play a pivotal role in facilitating the effective delivery of medical services to the community.

UI Walkthrough

Login Page - Start Page

WELCOME



Select Login

Patient

Staff

This is the login page for Arizona Medical. We can see two buttons, directing users to the Patient or Staff login pages respectively.

Login Page - Staff



Select User

Doctor

Nurse

We enter this view when the user selects the staff login. We can see two buttons, directing staff to the doctor's portal or nurse's portal respectively.

Patient Portal - Login



Patient Login

First Name:	
Last Name:	
Date of Birth (MM/DD/YYYY):	/ /

Confirm

New here?

Create Account

We enter this view when the user selects the patient login. We can see five fields where information can be entered: the patient's first name, their last name, and their date of birth split into month, day, and year. After entering all of the fields the user presses the confirm button to proceed to the patient portal. If the entered information is not in the database, an error message saying "Login not recognized. If you're new here, click 'Create Account'" is displayed above the confirm button. The Create Account button directs the user to the account creation page.

Patient Portal - Account Creation



Create Your Account

First Name:	
Last Name:	
Date of Birth (MM/DD/YYYY):	/ /

Confirm

Back

We enter this view when the user selects “Create Account” on the Patient Portal login page. We can see five fields where information can be entered: the patient’s first name, their last name, and their date of birth split into month, day, and year. After entering all of the fields the user presses the confirm button to create an account and proceed to the patient portal. If the entered information is already in the database, an error message saying “Account already registered” is displayed above the confirm button. The back button directs the user back to the patient portal login page.

Patient Portal - Menu



Welcome!

What would you like to do today?

Change Patient Information

View Previous Visits

Messages ¹

Logout

Once the patient logs into the application, they will be met with this home menu, where they have multiple options available to them, including viewing and changing their information, seeing past visits, and the ability to message staff members about certain things. The red number on the top right of the “Messages” button indicates the amount of new messages the patient has received since last looking at the message board. Once the patient is finished with their task, they also have the option to logout of their account to prevent others from viewing their information if they are sharing a computer with someone or using a public computer.

Patient Portal - Patient Information



Contact Information:

Phone Number:

xxx - xxx - xxxx

Insurance Information:

Insurance Provider

Pharmaceutical Information:

Pharmacy Provider

Update Information

Exit

This view shows the patient their information that is on file. Here, the patient is able to look over the information that they have provided and can make any changes to it if there are any mistakes or changes since the last time they checked this screen. Once the patient is satisfied with the information provided, they can click on the “Update Information” button to save their changes. The “Exit” button returns the patient to the home screen.

Patient Portal - Previous Visits



Select Visit to view: Visit Summary:

Visit Date ▼

Exit

This view shows a dropdown containing all of the patient's previous visits by the date of the visit. When the user selects a previous visit from the dropdown menu, the summary of the visit is displayed to the right. The user can then press the exit button to return to the menu when they are done.

Nurse Portal - Menu



Welcome!

What would you like to do today?

New Visit Form

Messages ²

Logout

Once the nurse logs into the application, they will be met with a home screen similar to that of the patient, but with different options available. The message and logout feature will still be available to the nurse, but they will have a “New Visit Form” button instead of the other features a patient has. This directs the nurse to the New Visit Form screen, where they can enter and view patient information for the next time the patient visits.

Nurse Portal - New Visit Form



Vitals:

Weight:
<Loaded Data> lbs

Height:
<Data> " feet/inches

Body Temperature:
<Data> F°

Blood Pressure:
<Data>

*Patient must be over the age of 12 to enter vitals

Submit for Physical

Exit

Patient Credentials:

First Name: <Text loaded from relevant .txt file>

Last Name: <Text loaded from relevant .txt file>

Date of Birth (MM/DD/YYYY): <Data> / <Data> / <Data>

Nurse's Notes:

Known Allergies:

<Text loaded from relevant .txt file>

Other Health Concerns:

<Text loaded from relevant .txt file>

Patient's Previous History:

Previously prescribed medications:

<Text loaded from relevant .txt file>

Immunization History

<Text loaded from relevant .txt file>

Previous Health Issues:

<Text loaded from relevant .txt file>

This view shows the nurse the information needed for the patient when they are having a new visit to the office. The nurse is able to see previous information of the patient and can also enter notes which will update this information once the form is submitted. The nurse should enter relevant information upon consultation. If the patient does not already have an account, a new one is created upon clicking of the button "Submit for Physical." The "Exit" button will return the nurse to the home screen.

Doctor Portal - Menu



Welcome!

What would you like to do today?

Patient Visit Form

View Patient Records

Messages 5

Logout

Once the doctor logs into the application, they are met with this home screen. The doctor is also able to view messages between them and patients as well as the ability to logout of their account once they are finished with their job. The doctor can view the Patient Visit Form that was created by the nurse where they can read the patient's information of the current visit that was taken by the nurse. The "View Patient Records" button redirects the doctor to a page where they can view previous records of certain patients which includes summaries of previous visits.

Doctor Portal - Patient Visit Form



Vitals:

Weight:	<Loaded Data>	lbs
Height:	<Data>	feet/inches
Body Temperature:	<Data>	F°
Blood Pressure:	<Data>	

*Patient vitals available for ages older than 12

Conduct Exam

Exit

Patient Credentials:

First Name:	<Text loaded from relevant .txt file>				
Last Name:	<Text loaded from relevant .txt file>				
Date of Birth	<Data>	/	<Data>	/	<Data>
(MM/DD/YYYY):					

Nurse's Notes:

Known Allergies:
<Text loaded from nurse's relevant .txt file>
Other Health Concerns:
<Text loaded from nurse's relevant .txt file>

Patient's Previous History:

Previously prescribed medications:
<Text loaded from relevant .txt file>
Immunization History
<Text loaded from relevant .txt file>
Previous Health Issues:
<Text loaded from relevant .txt file>

This view displays all of the patient's information to the doctor, and is similar to the visit form used by nurses. After reviewing the information as well as the nurse's notes after screening the patient, the doctor can choose to conduct a physical exam, taking them to the physical exam form.

Doctor Portal - View Patient Record



Select Patient:

Select Visit to view:

Exit

Visit Summary:

This view shows two dropdowns. First we select a patient from the first dropdown containing a list of patients by name and date of birth, which updates the second dropdown with a list of visit dates for that patient. Then we select a visit date from the second dropdown. After selecting a visit date, the summary of the visit is displayed to the right. We can then press the exit button to return to the menu when we are done.

Doctor Portal - Conduct Exam



Doctor's Exam:

Physical Exam:

Other Notes:

Update Summary

Exit

Prescription:

Prescription Script

Current Patient's Pharmacy

<Loaded from relevant .txt file>

Signature

Send Prescription

This view is opened after the doctor selects “Conduct Exam”, a physical exam in the patient visit form. The doctor writes a summary of the exam and its results in the designated space on the left and any miscellaneous information on the right. After finishing his/her report, they enter it into the system by clicking “Update Summary”. The doctor can prescribe the patient new medications in their treatment plan, which is authorized by their signature in the bottom space. Once the prescription, the patient’s pharmacy, and the doctor’s signature are finalized, they can update the patient’s prescription in the system by clicking “Send Prescription”. The prescription script will also be included in the “Visit Summary” text for its correlating visit, which both patients and doctors can view.

Messaging Board - Doctor & Nurse (Staff)



Select Patient:

Message Board:

Message:	Inbox: [Patient]: //Message Here [Doctor <u>or</u> Nurse]: //Message Here ... [Patient]: //Message Here [Doctor <u>or</u> Nurse]: //Message Here Patient Phone Number: <Phone Number loaded from .txt file>
Send Message	Exit

This view is opened when the doctor or nurse selects messages from their respective portal landing page. It contains a dropdown where the staff member selects a patient from the list by their name and date of birth, with patients who have unread messages filtered to the top of the list. Selecting a patient loads the message inbox for that patient, as well as the patient's phone number. The staff member can then type a message to the patient in the box to the left, and press "Send Message" to deliver the message to the patient's inbox, which will be displayed from either Nurse or Doctor depending on who is logged in.

Messaging Board - Patient



Message Board:

Message:	Inbox:
	[Patient]: //Message Here [Doctor <u>or</u> Nurse]: //Message Here ... [Patient]: //Message Here [Doctor <u>or</u> Nurse]: //Message Here

Send Message

Exit

This view is opened when the patient selects messages from the patient portal landing page. The patient's inbox is displayed on the right, containing all messages sent between the patient and the doctors/nurses. The patient can write a message in the box to the left, and press send to deliver the message to the staff members' inbox.

Credit Sheet

Team Member Name	Contributions
John Bostater	UI Design and UI Walkthrough (20%)
Tristan Andrade	UI Design and UI Walkthrough (20%)
Austin Mayhew	UI Design and UI Walkthrough (20%)
Ryan Clark	UI Design and Project Overview (20%)
Juan Rosas Jr.	UI Design and UI Walkthrough (20%)